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Fecal microbiota transplantation for refractory chronic graft-versus-host disease after allogeneic hematopoietic cell transplantation: a pilot open-label, non-placebo-controlled study

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Abstract

Background Dysbiosis of the intestinal microbiota plays a crucial role in the initiation and development of graft-versus-host disease (GVHD). Fecal microbiota transplantation (FMT) has been reported to be effective for refractory acute GVHD; however, whether FMT is effective for refractory chronic GVHD (cGVHD) remains unknown.

Methods To investigate the efficacy and safety of FMT for refractory cGVHD and the underlying mechanism, 12 patients with refractory cGVHD received FMT via colonoscopy, and the response was evaluated at 12 weeks after FMT.

Results Among the 12 patients who underwent FMT, 1 patient achieved a complete response, and 5 patients achieved a partial response. Patients with refractory cGVHD presented lower α diversity and higher abundance of *Escherichia-Shigella* and *Enterobacteriaceae*. FMT increased gut microbial diversity, increased the abundance of short-chain fatty acid (SCFA)-producing bacteria, and decreased the abundance of *Escherichia-Shigella* and *Enterobacteriaceae* in responder patients. Moreover, it increased SCFA levels in fecal samples from the responder group and promoted the expansion of peripheral CD4⁺CD127⁻ regulatory T (Treg) cells. Colon pathological examination revealed that CD4⁺ T and CD19⁺ B cell infiltration decreased and that CD4⁺ Treg infiltration increased after FMT.

Conclusions The results of the present study suggest that FMT is feasible and deserves further investigation for use in patients with refractory cGVHD.

Trial registration ClinicalTrials.gov (NCT06938165).

Keywords Allogeneic hematopoietic cell transplantation, Graft-versus-host disease, Fecal microbiota transplantation, Short-chain fatty acid, 16S rRNA sequencing

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Background

Allogeneic hematopoietic cell transplantation (allo-HSCT) is a curative procedure for many malignant hematologic diseases. Chronic graft-versus-host disease (cGVHD) remains a serious and prevalent long-term complication that can occur after allo-HSCT. Although marked progress has been made in graft-versus-host disease (GVHD) prophylaxis over the past few decades, cGVHD still occurs in 30–70% of recipients [1, 2]. Corticosteroids and calcineurin inhibitors together constitute the backbone of the standard first-line treatment for cGVHD, but the efficacy is only approximately 50% [3]. To date, although several therapies, including ruxolitinib, ibrutinib, axatilimab, and belumosudil, have been approved by the U.S. Food and Drug Administration (FDA) for refractory cGVHD, there is still no consensus regarding the standard second-line treatment [4–11]. Therefore, exploring new strategies for the treatment of cGVHD, particularly refractory cGVHD, without inhibiting the graft-versus-leukemia effect is urgent and crucial for improving patient outcomes after allo-HSCT [12–16].

The gut microbiota is essential for immune system development and maturation and plays a pivotal role in regulating host immunity. Dysbiosis of the gut microbiota is implicated in many autoimmune and inflammatory diseases, including GVHD [17–23]. Fecal microbiota transplantation (FMT) was first demonstrated to be safe and effective in patients with systemic lupus erythematosus (SLE) [24]. cGVHD shares similar clinical manifestations and mechanisms with SLE and other autoimmune diseases [25]. Growing evidence also suggests that gut microbiota dysbiosis occurs in GVHD patients [26, 27]. Although a few studies have demonstrated the effectiveness of FMT for acute GVHD (aGVHD), its therapeutic efficacy in treating cGVHD patients has rarely been reported. A study of FMT for eliminating antibiotic-resistant bacteria in patients with gastrointestinal GVHD, which included 2 extensive cGVHD patients, suggested that FMT might effectively eliminate gastrointestinal pathogens and alleviate cGVHD symptoms [28]. In this prospective, single-center study, we evaluated the efficacy and safety of FMT for treating refractory cGVHD and explored the underlying mechanism.

Methods

Study design

This was a single-center intervention study to assess the efficacy and safety of FMT for patients with refractory cGVHD who underwent allo-HSCT. The patients enrolled in this study were aged 18 to 65 years and were resistant to first- and second-line systemic therapy, which included corticosteroids and calcineurin inhibitors (CNIs) alone or in combination with second-line drugs,

for more than 2 weeks, had persistent cGVHD without improvement after one month of second-line treatment, or were unable to be off of steroids or other immunosuppressive drugs for 1 month [29]. The sample size was not formally calculated because this study aimed only to assess the efficacy and safety of FMT in a specific cohort of patients with refractory cGVHD. The study received institutional review board approval and was conducted in accordance with the Declaration of Helsinki. We followed the CONSORT statement when writing our manuscript (Additional file 1: CONSORT checklist).

FMT was administered intracolonicly once every 4 weeks for a total of 2 times, and the efficacy was evaluated at 12 weeks after the first FMT treatment. The cGVHD grade and response were assessed using the 2014 National Institutes of Health (NIH) Consensus Criteria [29]. The response was classified as a complete response (CR), a partial response (PR), or no response (NR). CR was defined as the resolution of all reversible manifestations in each organ or site of cGVHD involvement. PR was defined as improvement in at least 1 organ or site with a minimal score reduction of 1. NR was defined as no improvement in any organ or site of cGVHD lasting for 1 month, cGVHD disease progression or inability to taper steroids or other immunosuppressive drugs for 1 month. The primary outcome was the overall response rate (ORR) at 12 weeks after FMT. The secondary outcomes included donor microbiota engraftment, the response rate of the organ system, the number and proportion of refractory cGVHD patients who achieved the best CR or PR, overall survival (OS), the estimated colonic pathology score, and the change in the intestinal bacterial profile, which was analyzed by 16S rRNA gene sequencing at 12 weeks after FMT. Adverse events (AEs) were recorded within 12 weeks after FMT according to the National Cancer Institute Common Terminology Criteria for Adverse Events, version 4.0 [30].

FMT

The FMT donors were healthy volunteers who did not have a history of infectious diseases or inflammatory conditions, such as inflammatory bowel disease or gastrointestinal malignancies. Fecal matter samples from healthy donors were provided by Shenzhen Xbiome Biotechnology Co., Ltd. [31, 32]. The fecal microbiota suspensions for FMT were prepared as follows: (i) the crude stool samples were diluted with 500 ml of 0.9% normal saline; (ii) the fecal microbiota suspensions were extracted automatically by an intestinal bacteria extractor; (iii) the extracted fecal microbiota suspensions were centrifuged at 4000 rpm for 10 min, and the supernatants were discarded; and (iv) the precipitates were precisely diluted with 75% normal saline and 25% pharmaceutical-grade

glycerol and stored at -80°C for further application. Prior to delivery, the frozen samples for FMT were slowly thawed in a water bath ($+37^{\circ}\text{C}$) for 60 min before being mixed with 240 ml of sterile saline. To perform the FMT procedure, 300 ml of the fecal microbiota suspension was infused over 2–3 min into the cecum of the patient. Colon biopsies were collected via colonoscopy. The tissues were fixed with 4% polyformaldehyde at 4°C for 24 h and embedded in paraffin. Two stool samples were collected from one donor (D1-1, D1-2), and three stool samples were collected from another donor (D2-1, D2-2, D2-3). The FMTs of patients #1, #5, #6, and #9 were performed using the D1-1 and D1-2 samples. Two stool samples were collected from one donor (D1), and three stool samples were collected from another donor (D2). FMT for patients #1, #5, #6, and #9 was performed using the D1 fecal microbiota suspension. FMT for patients #2, #3, #4, #7, #8, #10, #11, and #12 was performed using D2 fecal microbiota suspension.

Fecal microbial composition analysis

Fecal samples were collected from the recipients before FMT and 12 weeks after FMT [22]. The samples were stored at -80°C until RNA extraction. After RNA extraction, bacterial 16S rRNA was successfully detected in all of the samples by polymerase chain reaction (PCR) using the following universal bacterial primers (16S V4-V5): 515F: 5'-GTGCCAGCMGCCGCGGTAA-3'; 926R: 5'-CCGTCAATTCMTTGTAGTTT-3'. After purification, the pooled libraries were sequenced by 2×300 bp paired-end sequencing on the MiSeq platform (Tiny Gene, Shanghai, China) using the MiSeq v3 Reagent Kit (Illumina). Microbiome bioinformatics was performed with QIIME2 2019.4. The raw sequences were demultiplexed using the Demux function, and the primers were cut by the Cutadapt plugin. The DADA2 plugin was used to generate amplicon sequence variants (ASVs) at a 97% similarity threshold. Quality filtering, denoising, combination, and removal of chimeric sequences were performed by the DADA2 pipeline. Taxonomy was assigned to ASVs using the classify-sklearn naïve Bayes taxonomy classifier in the feature-classifier plugin against the Silva_138.2 16S rRNA database at 70% confidence [33–35]. The alpha diversities were determined by the phyloseq package in R (version 4.1). The similarity of the microbial community structure between samples was examined using principal coordinate analysis (PCoA) on the basis of the Bray–Curtis distance algorithm.

Immunohistochemistry

Tissue sections of $4\text{-}\mu\text{m}$ thickness were prepared from formalin-fixed and paraffin-embedded blocks. The tissue sections were heated at 65°C for 1 h, followed by

deparaffinization in xylene and rehydration in an ethanol gradient. Blocking and quenching of endogenous peroxidase activity were performed with 3% hydrogen peroxide for 25 min at room temperature. Antigen retrieval was performed by using ethylene diamine tetraacetic acid (EDTA) buffer (MVS-0098, Maixin Biotech). The sections were subsequently blocked with 1% bovine serum albumin (BSA) solution for 60 min at room temperature. Next, the sections were incubated with antibodies against human CD4 (1/300, RMA-0620, Maixin Biotech), CD8 (1/300, RMA-0514, Maixin Biotech), CD11b (1/300, ab133357, Abcam), E-cadherin (1/300, MAB-0738, Maixin Biotech), Foxp3 (1/100, ab20034, Abcam), and CD19 (1/300, MAB-0705, Maixin Biotech). The slides were then incubated with horseradish peroxidase (HRP)-conjugated anti-mouse (Kit-0014, Maixin Biotech) or HRP-conjugated anti-rabbit (Kit-0015, Maixin Biotech) secondary antibodies for 1 h at room temperature. Antibody staining was visualized using an H_2O_2 /diaminobenzidine (DAB) reaction kit (MaxVision III Ultra DAB, Kit-0038, Maixin Biotech). After counterstaining with hematoxylin (Wako), the sections were dehydrated in ethanol and xylene and then sealed with glass coverslips using neutral balsam.

The stained slides were scanned using a slide scanner (EVIDENT VS200) at $40\times$, and the digital imaging results were analyzed using QuPath-v0.2.3. software. In the lamina propria layer of the sections, we randomly chose five different representative regions containing approximately 50–100 cells to measure their DAB staining optical density (OD) automatically via the software.

Analysis of immune cells and SCFA levels

Peripheral blood mononuclear cells (PBMCs) from recipients were collected and stained with FITC-conjugated mouse anti-human CD3 (SK7), APC-conjugated anti-human CD4 (SK3), PE-conjugated anti-human CD8 (SK1), APC-Cy7-conjugated anti-human CD25 (2A3), and PerCP-conjugated anti-human CD127 (HIL-7R-M21) antibodies. Samples were acquired using a BD FACSCanto II (BD Biosciences) and analyzed with FlowJo software (Tree Star, Ashland, OR). The plasma samples from the recipients were stored at -80°C and prepared for metabolomic analysis according to the manufacturer's instructions (Majorbio Bio-Pharm Technology Co., Ltd., Shanghai). SCFA analysis was performed using ultra-performance liquid chromatography coupled with tandem mass spectrometry (LC/MS) on a Thermo UHPLC-Q Exactive HF-X system.

Establishment of the bone marrow transplant (BMT) model

The animal experiments were performed in accordance with the ARRIVE guidelines (Additional file 2).

MHC-mismatched mice (C57BL/6→BALB/c) were used to establish the murine cGVHD model. Wild-type BALB/c and C57BL/6 female mice (8–10 weeks old; body weight: 20 ± 2 g) were purchased from Charles River Laboratories (Beijing Vital River Laboratory, China), and no more than 5 mice were bred per cage with corn-cob bedding at the animal facilities of Nanfang Hospital (Guangzhou, China). The mice were housed in a specific pathogen-free (SPF) environment at a constant temperature (24 ± 2 °C) with a 12-h dark/light cycle. The animals had free access to a grain-based diet and water. The mice were acclimatized for one week before the experiment began. All procedures for the animal experiments were performed at the animal facilities of Nanfang Hospital (Guangzhou, China) in an SPF environment. The BALB/c recipients received 675 cGy lethal-dose irradiation for BMT [4, 36]. Single-cell suspensions were prepared by flushing bone marrow (BM) from donor femurs and tibias and gently breaking down the spleens through sterile mesh filters. The irradiated recipient mice received BM cells via intravenous injection into the lateral tail vein, either alone or in combination with donor splenocytes. Weight- and size-matched mice were randomly divided into the following groups: the BM alone (BMA) group, the singly housed group, and the cohoused groups. After BMT, the mice were monitored twice a week for weight loss and other clinical signs of GVHD. Scores were recorded for five clinical parameters associated with cGVHD: weight loss, posture, activity, fur texture, skin integrity, diarrhea, and swollen or partially closed eyes. Each mouse was scored from 0 to 2 in each category, with a maximum score of 14 representing cGVHD. The mice were euthanized if the score was greater than 10 for 2 continuous measurements.

To evaluate the effect of the donor gut microbiota on cGVHD, age- and sex-matched female B6 donor littermates were housed with cGVHD recipients in our study. At 30 days after transplantation, cGVHD mice were housed either alone or with healthy B6 donor mice at a 1:1 ratio for 4 weeks, after which target organs were collected from the recipient mice for pathological evaluation and further analysis. The corresponding author was aware of the group allocations at all stages of the experiment.

Statistical analysis

The microbiome sequence data analyses and calculations were performed using the QIIME2 platform and phyloseq package in R (version 4.1). All the statistical analyses were performed using GraphPad Software (Prism Version 6.0; GraphPad Software, San Diego, CA) or SPSS version 22.0

(SPSS, Chicago, IL). There were no missing data in the study. The ORR and adverse events are considered categorical variables, and Fisher's exact test was performed to compare categorical variables. Comparisons of two groups were assessed by an unpaired two-tailed Student's *t* test or the Mann–Whitney test. One-way ANOVA with Dunnett's correction was performed for multiple group comparisons. Linear discriminant analysis (LDA) effect size (LEfSe) was performed to identify differentially abundant taxa across groups using the default parameters. Donor microbiota engraftment was measured by the community of unique donor amplicon sequence variants (ASVs). These ASVs represent the microorganisms present in the donor feces and patient post-FMT feces but not in the patient pre-FMT feces. This community was used as the algorithm surrogate for measuring donor microbiota engraftment, similar to a previous study [37–39]. The Jaccard index was used to calculate the similarity of the gut microbial composition of the receiver to that of their fecal donors [40]. The data are expressed as the mean $\pm$ standard deviation (SD) or mean $\pm$ standard error of the mean (SEM). On the basis of preliminary experiments and published resources for in vivo experiments, a sample size of $n=5$ per group was determined to be sufficient to detect an effect size of 0.775, with a significance level (α) of 0.05 and a statistical power of 0.8 ($1-\beta$). Tests of significance were 2-sided, with a *p* value of 0.05 or less.

Results

Patient characteristics

A total of 12 patients were enrolled between April 2020 and February 2022 for this study, and the last follow-up was June 2024. The transplantation-related and disease-related characteristics of the patients and the types of GVHD prophylaxis used are summarized in Table 1. Ten patients developed de novo cGVHD, and the other 2 patients developed cGVHD after donor lymphoid cell infusion. In addition to receiving corticosteroids, all patients had been treated with at least one other form of systemic immunosuppression (cyclosporin, cyclophosphamide, thalidomide, ruxolitinib, or tacrolimus) at the time of receiving FMT. The enrolled patients had received a median of 2 prior treatments and were refractory to their last cGVHD treatment before study enrollment. The median duration of the last therapy for cGVHD treatment was 2 months (range, 1–4 months). The median time from transplantation to cGVHD onset was 7 months (range, 4–22 months). The median time from cGVHD diagnosis to enrollment in the study was 13 (8–25) months. Among these patients, 2 patients (16.7%) had moderate cGVHD, and 10 patients (83.3%) had

Table 1 Patient characteristics

ID	Age	Sex	Primary disease	Stem cell source	GVHD prophylaxis	cGVHD-involved organs	Overall cGVHD severity	Survival	Lines of therapy prior to FMT treatment	Duration of prior therapy (range), month
P01	27	Male	ALL	PBSC	CsA + MTX + MMF	Oral/Eyes/Joints	Severe	Live	2	3
P02	45	Female	ALL	PBSC	CsA + MTX + MMF	Lungs	Severe	Live	2	2
P03	36	Female	MDS	PBSC	CsA + MTX + MMF	Eyes/Lungs	Severe	Live	2	3
P04	65	Female	AML	BM + PBSC	CsA + MTX + MMF	Skin	Severe	Live	2	2
P05	18	Female	AML	PBSC	CsA + MTX + MMF + ATG	Lungs	Severe	Live	2	4
P06	31	Male	ALL	PBSC	CsA + MTX + MMF	Liver/Skin	Moderate	Live	1	2
P07	57	Male	MDS	BM + PBSC	CsA + MTX + MMF + ATG	Eyes/Liver/Lungs/Skin	Severe	Dead	2	3
P08	52	Male	MDS	BM + PBSC	CsA + MTX + MMF + ATG	Eyes/Kidney	Severe	Live	3	2
P09	29	Male	AML	PBSC	CsA + MTX + MMF	Skin/Joints/Eyes/Liver/Lungs	Severe	Dead	2	2
P10	45	Female	ALL	BM + PBSC	CsA + MTX + MMF + ATG	Oral/Liver	Severe	Live	1	2
P11	34	Male	NHL	PBSC	CsA + MTX + MMF + ATG	Skin	Moderate	Live	3	1
P12	51	Female	AML	BM + PBSC	CsA + MTX + MMF + ATG	Skin/Liver	Moderate	Live	2	2

ALL acute lymphoblastic leukemia, AML acute myeloid leukemia, ATG antithymocyte globulin, BM bone marrow, cGVHD chronic graft-versus-host disease, CsA cyclosporine A, FMT fecal microbiota transplantation, HSC hematopoietic stem cell transplantation, GI tract gastrointestinal tract, MDS myelodysplastic syndrome, MMF mycophenolate mofetil, MTX methotrexate, NHL non-Hodgkin lymphoma, PBSC peripheral blood stem cell

severe cGVHD. Among the patients with severe cGVHD, lung involvement occurred in 5 patients (55.6%), with frequent coinvolvement of the eyes (4 patients, 44.4%), skin (4 patients, 44.4%), and liver (4 patients, 44.4%). Joints and the oral cavity were affected 2 patients (22.2%), whereas kidney involvement was observed in 1 patient (11.1%). Seven of 9 patients (77.8%) showed involvement

of two or more organ systems. Among the patients with moderate cGVHD severity, isolated skin involvement occurred in 2 patients (66.7%), and combined skin and liver involvement occurred in 1 patient (33.3%). Among all 12 enrolled patients, 10 (83.3%) had cGVHD involving the GI tract, as confirmed by histopathological diagnosis (Additional file 3: Fig. S1).

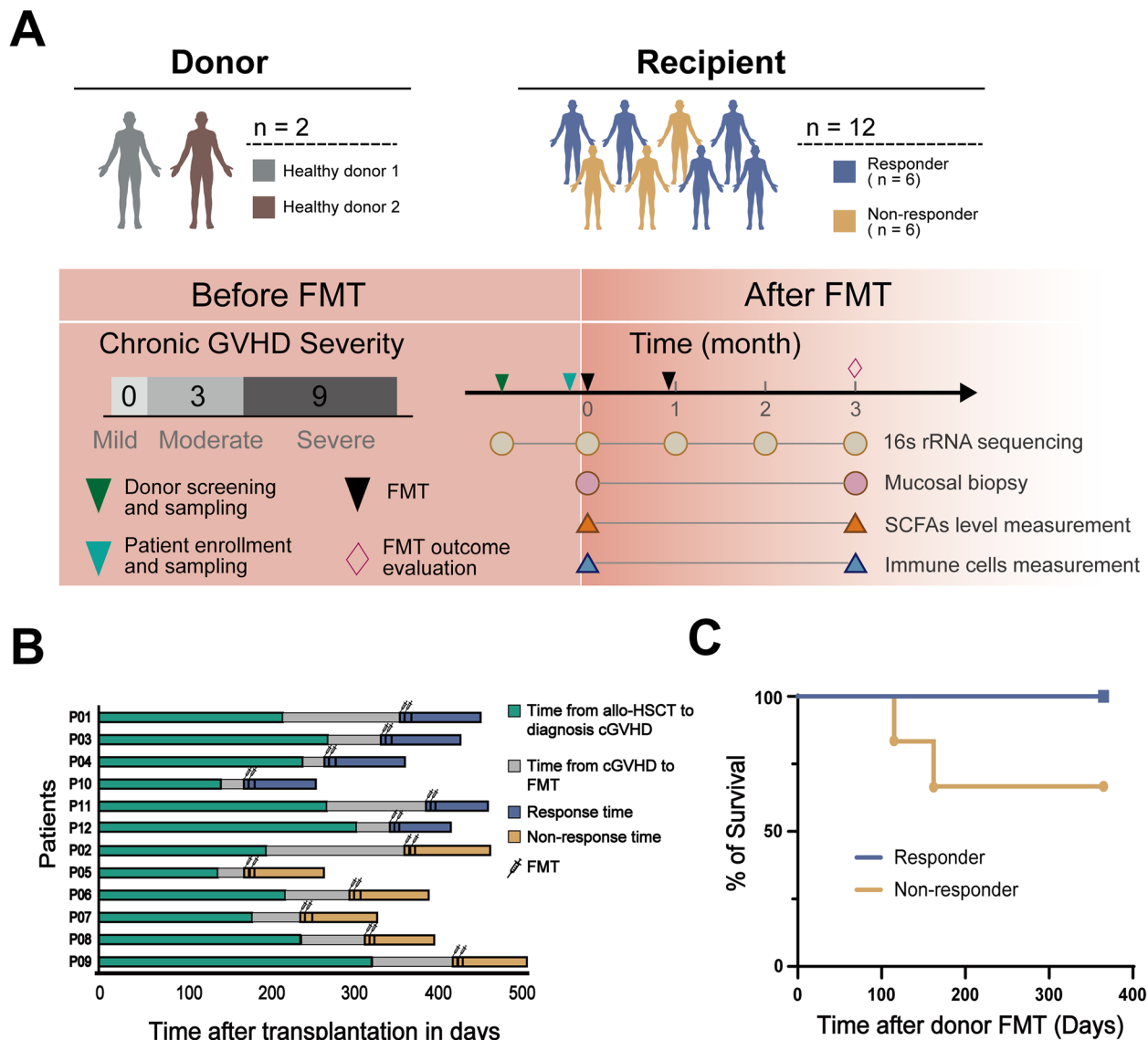


Fig. 1 Graphical illustration of the study design and FMT outcome for patients with refractory chronic GVHD in this study. **A** Outline of the study design and FMT outcome. The donor was thoroughly screened with questionnaires for behavioral risk factors, and blood, urine, and stool laboratory tests were performed to avoid samples with potentially transmissible pathogens, including *Streptococcus*, *Escherichia coli* pathotypes, and *Mycobacterium tuberculosis*, and COVID-19 testing was conducted according to international guidelines [54]. All participants received two rounds of FMT. The disease characteristics and FMT outcomes of the twelve recipients are depicted in the upper and bottom panels. 16S rRNA sequencing was performed monthly before and after FMT until the FMT outcome was evaluated. Colonic mucosal biopsy, SCFA levels in the plasma and stool, and immune cells in the circulating blood and colon were measured before and after FMT. **B** Timeline of FMT conducted on patients according to the date of allo-HSCT. **C** Kaplan–Meier analysis of the survival time between the responder group and the nonresponder group followed by FMT. FMT fecal microbiota transplantation, SCFAs short-chain fatty acids, cGVHD chronic GVHD; * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, and **** $p < 0.0001$

Efficacy and safety

To assess the efficacy and safety of FMT for allo-HSCT recipients who are refractory to cGVHD, the twelve participants received health donor FMT treatment. The outline of the study is provided in Fig. 1A. Of the 12 patients, after two FMTs, 1 patient achieved a CR, 5 patients achieved a PR, and 6 patients continued to exhibit NR (Fig. 1B). Among these patients, all the responders survived, whereas 2 nonresponders died at 115 days and 162 days after FMT (100% vs. 66.7%, hazard ratio (HR)=0.167, 95% CI: 0.0102–2.718, $p=0.208$) (Fig. 1C). In the responder group, one patient achieved a significant response after the first FMT, leading to the tapering of immunosuppressant drug therapy at 12 weeks and complete discontinuation within 3 months after FMT. Five patients experienced improvement in cGVHD symptoms and began tapering immunosuppressant drug therapy after two FMTs. In the nonresponder group, FMT did not result in any changes in clinical symptoms, and patients were unable to taper their immunosuppressant drugs; 2 patients experienced sustained dyspnea and exacerbated skin scleroderma, which was considered indicative of cGVHD progression. The median time from cGVHD diagnosis to FMT in the responder group was 12 (range, 4–20 weeks), whereas the median time from FMT in the nonresponder group was 24 (range, 16–32 weeks).

To assess whether the FMT response varied across cGVHD-targeted organs, the response rate was evaluated across all affected organs. In the responder group, organ-specific analyses revealed an ORR of 50% (3/6) in the skin, 40% (2/5) in the eyes, 20% (1/5) in the lungs, 40% (2/5) in the liver, 100% (2/2) in the oral mucosa, and 50% (1/2) in the joints. No response was observed in the patient with kidney involvement (0/1) (Table 2). Chi-square tests revealed no statistically significant difference in FMT efficacy for various cGVHD-targeted organs (skin: $p=1.000$, Pearson chi-square=0.000; eyes: $p=0.558$, Pearson chi-square=0.343; lungs: $p=0.079$, Pearson chi-square=3.086; liver: $p=0.558$, Pearson chi-square=0.343; oral mucosa: $p=0.121$, Pearson chi-square=2.400; joints: $p=1.000$, Pearson chi-square=0.000) (Additional file 3: Table S1).

Analysis of the microbiota composition in the donor bacterial solution revealed that the microbial profiles of Donor 1 and Donor 2 were similar across different taxonomic levels (Additional file 3: Fig. S2). Four patients (P01, P05, P06, and P09) underwent FMT with fecal microbiota suspension from Donor 1, which had relatively high abundance of *Faecalibacterium*, *Prevotella*, and *Klebsiella*. The outcomes for these patients were as follows: 1 patient achieved a PR, and 3 patients had NR at the endpoint. Eight patients (P02, P03, P04, P07, P08, P10, P11, and P12) underwent FMT with fecal microbiota suspension from Donor 2, which had high abundance

Table 2 Summary of ORRs of various cGVHD-involved organs after FMT

Organ, n(%)	FMT responder group (n=6)	Total FMT cohort (n=12)
Skin (n=6)		
CR	2/3 (66.7%)	2/6 (33.3%)
PR	1/3 (33.3%)	1/6 (16.7%)
ORR	3/3 (100%)	3/6 (50.0%)
Eyes (n=5)		
CR	1/2 (50.0%)	1/5 (20.0%)
PR	1/2 (50.0%)	1/5 (20.0%)
ORR	2/2 (100%)	2/5 (40.0%)
Lungs (n=5)		
CR	1/1 (100%)	1/5 (20.0%)
PR	0/1 (0%)	0/5 (0%)
ORR	1/1 (100%)	1/5 (20.0%)
Liver (n=5)		
CR	1/2 (50.0%)	1/5 (20.0%)
PR	1/2 (50.0%)	1/5 (20.0%)
ORR	2/2 (100%)	2/5 (40.0%)
Oral mucosa (n=2)		
CR	1/2 (50.0%)	1/2 (50.0%)
PR	1/2 (50.0%)	1/2 (50.0%)
ORR	2/2 (100%)	2/2 (100%)
Joints (n=2)		
CR	0/1 (0%)	0/2 (0%)
PR	1/1 (100%)	1/2 (50.0%)
ORR	1/1 (100%)	1/2 (50.0%)

ORR overall response rate = the proportion of subjects who achieved complete remission (CR) or partial remission (PR); FMT fecal microbiota transplantation

of *Subdoligranulum*, *Ruminococcus*, and *Dorea*. At the end of the study, one patient achieved a CR, 4 patients achieved a PR, and 3 patients had NR.

The most frequent AEs are provided in Table 3. Patients tolerated FMT well without treatment-emergent AEs (TEAEs) within 6 h of FMT. Most AEs were grade 1 or 2, with the most common being bloating, nausea and diarrhea. No severe AEs (grade ≥ 3 events) were reported in this study. None of the subjects developed a bacterial, viral or fungal infection within 2 weeks after FMT.

FMT ameliorates microbiota dysbiosis in patients with refractory cGVHD

Next, we compared the microbiota of cGVHD patients with that of healthy donors by 16S rRNA analysis. Principal coordinate analysis (PCoA) revealed that the microbiota composition significantly differed between patients with refractory cGVHD and healthy donors; however, FMT altered the microbiota composition of

Table 3 Adverse events

TEAE, n (%)	FMT (n = 24)	
AEs, n(%)		SAEs, n(%)
Bloating	12 (50)	Intestinal Infection
Bloating	4 (17)	Sepsis
Diarrhea	2 (8)	Intestinal Perforation
Nausea	6 (25)	Bloody Stool
Dyspnea	1 (4)	Severe Pneumonia
Scleroderma	1 (4)	Death
Fever	0	
Vomiting	0	

TEAE treatment-emergent adverse event, FMT fecal microbiota transplantation, AEs adverse events, SAEs serious adverse events

patients to resemble that of healthy donors (Fig. 2A). The alpha-diversity Shannon index revealed that patients with refractory cGVHD had similar microbial diversity (Shannon index) to healthy donors ($p > 0.05$), but the abundance coverage estimator (ACE), Sobs and Chao1 indices were significantly lower, suggesting that refractory cGVHD patients have lower bacterial community richness than healthy donors do (Fig. 2B, $p < 0.001$). Compared with patients at baseline, patients at the beginning of FMT presented significantly elevated alpha diversity, Shannon indices and community richness indices (Fig. 2B). The patients with refractory cGVHD exhibited dysbiosis of the gut microbiota, and FMT significantly restored the intestinal microbial ecosystem, as indicated by an increased ratio of *Firmicutes* to *Bacteroidota* (healthy donors vs. patients with refractory cGVHD before vs. after FMT: 7.5141 vs. 3.0347 vs. 5.0801, Fig. 2C and Additional file 3: Fig. S3).

Next, we analyzed the microbiota composition of cGVHD patients at different hierarchical levels. At the genus level, the patients presented increased levels of *Enterobacteriaceae*, whereas the levels of *Ruminococcaceae*, *Acidaminococcaceae*, and *Coriobacteriaceae* were reduced. FMT decreased the level of *Enterobacteriaceae* and increased the levels of *Ruminococcaceae*, *Acidaminococcaceae*, and *Coriobacteriaceae*. At the species level, the abundance of *Escherichia-Shigella* and

beneficial taxa such as *Megamonas*, *Faecalibacterium*, *Subdoligranulum*, *Phascolarctobacterium*, and *Ruminococcus torques* decreased. FMT decreased the level of *Escherichia-Shigella* and increased the levels of *Megamonas*, *Subdoligranulum*, *Phascolarctobacterium*, and *Ruminococcus torques* (Fig. 2D). We assessed the donor microbiota engraftment by determining the fraction of unique donor taxa (donor origin) and the fraction of unique recipient taxa (recipient origin) in the follow-up stool samples from the study participants. In all patients, the proportion of the donor-derived gut microbiota increased after FMT (Fig. 2E; 0 week vs. 4 weeks post-FMT, $p = 0.031$; 0 week vs. 8 weeks post-FMT, $p = 0.033$; 0 week vs. 12 weeks post-FMT, $p = 0.015$). Although the donor-derived gut microbiota composition was not significantly different between responders and nonresponders, engraftment of donor fecal bacteria increased steadily in responders by 12 weeks post-FMT, whereas one nonresponder (P08) exhibited decreased engraftment of the donor-derived microbiota at this time point (Fig. 2F). The recipient-derived gut microbiota was unchanged after two rounds of FMT in all patients (Fig. 2G; 0 week vs. 12 weeks post-FMT, $p = 0.6118$). At baseline, the compositions of the recipient-derived gut microbiota in the responder and nonresponder groups were comparable (Fig. 2H, $p = 0.713$).

Differences in fecal microbiota composition and plasma metabolic and immune cell profiles between responders and nonresponders

To identify the microbes involved in the ability of FMT to ameliorate refractory cGVHD, we compared the composition of the microbiota between responders and nonresponders (Fig. 3A). The results revealed that the genera *Phascolarctobacterium* ($p = 0.018$; log-odds ratio -5.288 ; 95% CI: $-10.76-1.044$), *Faecalibacterium* ($p = 0.027$; log-odds ratio -2.081 ; 95% CI: $-4.069-0.4123$), *Parabacteroides* ($p = 0.030$; log-odds ratio -0.3479 ; 95% CI: $-1.6231-1.437$) and *Butyrivibrio* ($p = 0.017$; log-odds ratio -0.07721 ; 95% CI: $-0.1483-0.02235$) were more abundant in the responder group (Fig. 3B). The abundance of these strains of bacteria showed no change or even declined in the nonresponders.

(See figure on next page.)

Fig. 2 Gut microbiota dynamics in response to FMT in patients with refractory cGVHD. **A** PCoA of microbiota profiles demonstrating post-FMT alignment with donor samples. **B** Microbial diversity comparison among donors before FMT and after FMT in patients with refractory cGVHD. **C** The major bacterial components differed significantly before and after FMT. **D** Different levels of microbiota dysbiosis in refractory cGVHD patients before and after FMT. **E** The similarity of the gut microbiota of FMT recipients to that of their donor fecal resources before and after FMT. **F** The similarity of the gut microbiota of FMT recipients to that of their donor fecal resources in the FMT responder and nonresponder groups. **G** The similarity of the gut microbiota of FMT recipients to their original fecal microbiota before FMT. **H** The similarity of the gut microbiota of FMT recipients to their original fecal microbiota before FMT in the FMT responder and nonresponder groups. FMT fecal microbiota transplantation, cGVHD chronic GVHD, PCoA principal coordinate analysis; * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, and **** $p < 0.0001$

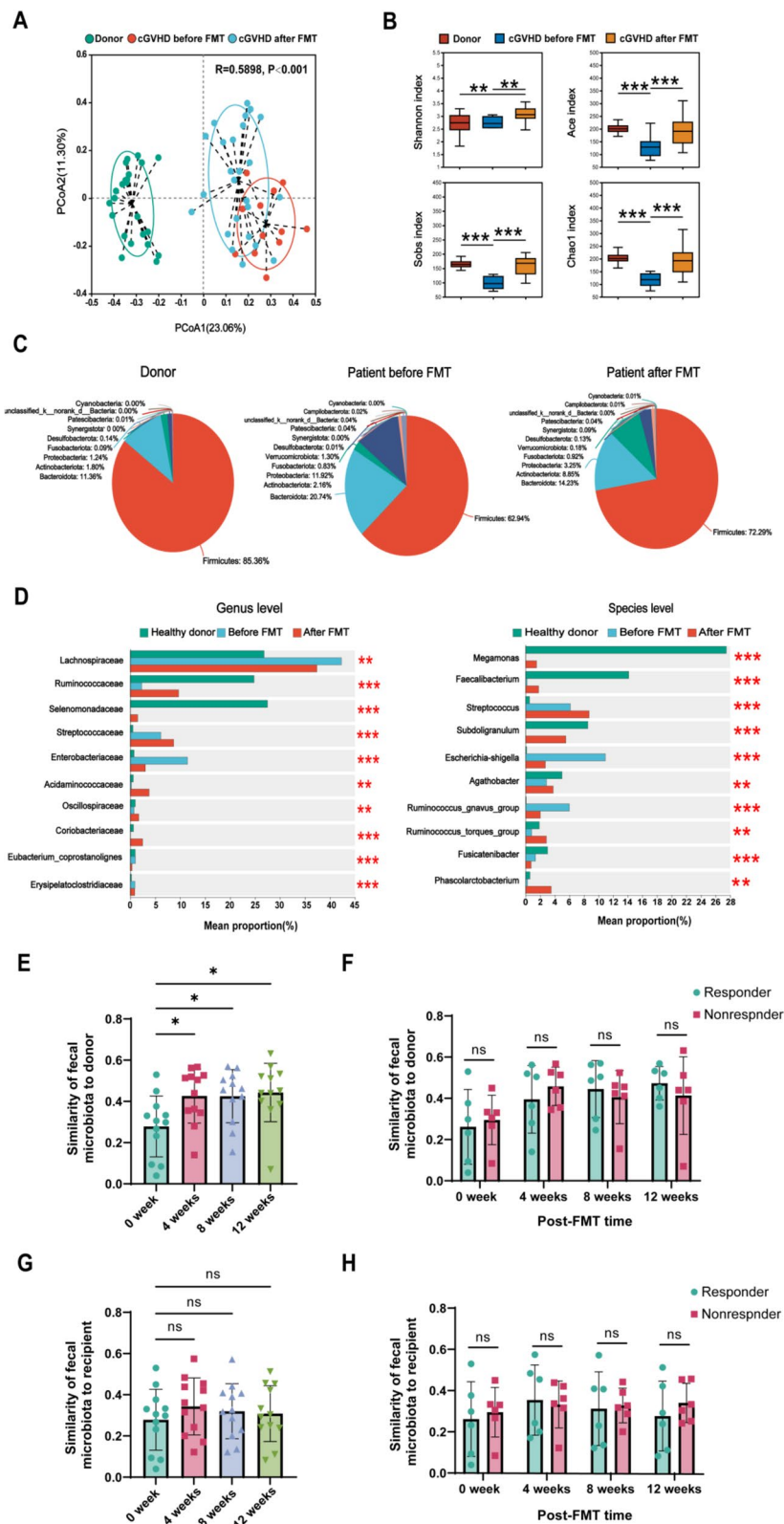


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Phascolarctobacterium, *Faecalibacterium*, and *Butyricimonas* species are well-known SCFA-producing bacteria [41–43]. Additionally, we detected other SCFA-producing bacteria enriched in responders, including *Faecalibacterium* (0.2835 vs. 2.3640, $p < 0.05$), *Dorea* (0.0948 vs. 1.7420, $p < 0.05$), *Subdoligranulum* (0.0161 vs. 5.8280, $p < 0.05$), *Coprococcus* (0.6978 vs. 1.0150, $p < 0.05$), and *Butyricicoccus* (0.1479 vs. 0.7997, $p < 0.05$) (Fig. 3C). LEfSe analysis showed similar findings, revealing that the abundance of *Phascolarctobacterium*, *Subdoligranulum*, and *Butyricimonas* was increased in responders (Fig. 3D).

FMT increased the levels of SCFAs, including acetic acid, butyric acid and valeric acid, in the fecal samples of the responders but not in those of the nonresponders (Fig. 3E). However, SCFA levels were not significantly increased in the plasma samples of responders (Fig. 3F). Among these significantly altered SCFAs, butyric acid effectively promoted Treg generation in patients with refractory cGVHD in vitro (Additional file 3: Fig. S4). Furthermore, the percentage of peripheral CD4⁺CD127[−] Treg (pTreg) cells significantly increased after treatment (Fig. 3G; $p = 0.0161$ and $p = 0.0318$, respectively), and Reg3 α (regenerating islet-derived protein 3) and ST2 (suppression of tumorigenicity 2) levels decreased (Fig. 3G). Histological analysis of colon biopsy samples revealed that FMT responders presented simultaneous decreases in the percentages of CD4⁺ T cells and CD19⁺ B cells, whereas the percentage of Foxp3⁺ cells increased, but the percentages of CD8⁺ T cells and CD11b⁺ myeloid cells remained stable (Fig. 4). Moreover, E-cadherin expression was partially increased in the colons of the responders (Additional file 3: Fig. S5).

Cohousing with healthy donor mice restored gut SCFA levels and Treg cell activity in recipient mice to alleviate chronic GVHD

To investigate whether the amelioration of gut microbiota dysbiosis contributes to the alleviation of cGVHD, we cohoused B6 cGVHD mice with BALB/c healthy donor mice for 4 weeks at a 1:1 ratio from day 28 after BMT and analyzed SCFA levels in fecal pellet and plasma samples and the percentages of splenic Treg cells in singly housed and cohoused cGVHD model mice (Fig. 5A). Age- and

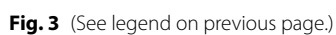
sex-matched singly housed cGVHD model mice served as controls. Cohousing with healthy mice reduced the severity of cGVHD, as reflected by the clinical scores (Fig. 5B). Compared with that in singly housed cGVHD model mice, damage to the gut and spleen was less severe in cohoused cGVHD model mice (Fig. 5C). The percentage of Foxp3⁺CD4⁺Treg cells was elevated in the spleens of cohoused cGVHD model mice (Fig. 5D). Moreover, the levels of SCFAs, particularly acetic acid and butyric acid, were increased in the feces (Fig. 5E). However, the measurement of SCFAs in the plasma of the mice revealed only a slight increasing trend (Fig. 5F).

Discussion

In this study, our results suggested that FMT is tolerated in patients with refractory cGVHD and that the underlying mechanism is associated with SCFA-producing bacteria and the immunoregulatory effect of SCFA metabolites on allogenic T cells. The ORR of FMT in refractory cGVHD patients at week 12 was 50.0%, with a CR rate of 8%, which is similar to that reported in patients with aGVHD or SLE [19, 24, 44]. The clinical outcome of FMT treatment is potentially beneficial for refractory cGVHD patients and is noninferior to that of standard first-line treatment, which consists of corticosteroids and calcineurin inhibitors for chronic GVHD, with a 30–50% response rate [3]. The ORR in our patient population was comparable with that of FDA-approved drugs that are used for refractory cGVHD, such as ibrutinib [7] (with the best ORR of 67% at 13.9 months), ruxolitinib [10] (with an ORR of 49.7% at week 24), and belumosudil [6] (with the best ORR of 74% by 6 months after treatment). Among the enrolled patients, four patients were refractory to ruxolitinib as second-line treatment at the time of receiving FMT treatment. FMT yielded an ORR for cGVHD of 75% in these ruxolitinib-refractory patients, with 3 of 4 patients achieving a PR when they remained on their previous ruxolitinib treatment. In this subpopulation that combined FMT with ruxolitinib treatment, the response rate led to clinically meaningful results and improvement in the affected organs, which suggests that FMT may help to overcome ruxolitinib resistance.

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Fig. 3 FMT attenuated refractory cGVHD by increasing the abundance of SCFA-producing bacteria. **A** Bacterial genus distributions in the responder and nonresponder groups across time points of FMT. **B** Changes in the abundance of *Phascolarctobacterium*, *Faecalibacterium*, *Parabacteroides*, and *Butyricimonas* over time in response to FMT. **C** Differential microbiota compositions at the genus level between patients with refractory cGVHD before and after FMT. **D** LEfSe analysis revealed significant changes in the SCFA-producing microbiota composition after FMT. **E** Targeted LC–MS measurements of fecal SCFA levels in the FMT responder and nonresponder groups. **F** Targeted LC–MS measurements of plasma SCFA levels in the FMT responder and nonresponder groups. **G** Inflammatory cytokine levels in the plasma and CD4⁺/CD8⁺/circulating CD4⁺ Treg cells in the responder group. **FMT** fecal microbiota transplantation, **SCFAs** short-chain fatty acids, **cGVHD** chronic GVHD, **LC/MS** liquid chromatography–mass spectrometry; * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, and **** $p < 0.0001$



The tolerance to FMT in our patients showed the feasibility of administering this treatment. No subjects discontinued FMT treatment because of treatment-related AEs. All reported AEs were manageable, with no serious AEs (grade ≥ 3) reported in this study. Currently, all FDA-approved refractory cGVHD treatment options are potent immunosuppressive drugs and, consequently, increase the risk of infection, such as the 10% pneumonia grade ≥ 3 TEAEs reported with axatilimab and 36% infection grade ≥ 3 TEAEs reported with ibrutinib. FMT may cause abdominal symptoms but exhibits reduced hematologic toxicity-related conditions, such as cytopenias, which are reported as the major serious AEs (grade 3) associated with ruxolitinib.

In our study, the most common AEs associated with FMT treatment, including bloating, nausea, and diarrhea, were mild. In the two nonresponders who experienced sustained nausea or worsening skin scleroderma after FMT treatment, these symptoms presented at the onset of cGVHD prior to study enrollment. No new rashes or other AEs were observed following FMT treatment. Therefore, these symptoms were considered unrelated to FMT treatment and were attributed to the progression of cGVHD.

A main concern is that administering a high load of donor-derived microorganisms to immunocompromised patients could increase the potential risk of infection. A systematic review of clinical studies involving immunocompromised patients revealed no additional risk when using donor FMT to treat *C. difficile* infection compared with its use in noninfected immunocompetent individuals [45]. In allo-HSCT recipients with aGVHD, studies have shown that FMT does not increase infection risk. For example, Kakihana et al. reported that FMT was used successfully to treat steroid-resistant gut aGVHD, and no additional infection risk was observed [46]. Similarly, Youngster et al. reported that FMT in allo-HSCT patients did not increase the risk of infection, supporting its safety in this vulnerable population [47]. In this work, our results revealed that multiple rounds of FMT were well tolerated by patients, with no evidence of FMT-related infections observed within 2 weeks after FMT. None of the patients received infection prophylaxis or were treated with antibiotics within one month before FMT. Only one participant (P01) had comorbid infections on

day 9 after receiving secondary donor FMT, and this patient received cefmetazole and cefoperazone-sulbactam for oral ulcers with signs of local purulent infection.

Currently, most research on the gut microbiota in GVHD is focused on aGVHD. Patients with aGVHD typically display a significant decrease in gut microbiota diversity, along with the overgrowth of pathogenic bacteria such as *Enterococcus* and *Streptococcus* and a lower level of SCFA-producing microbes (e.g., *Blautia* and *Faecalibacterium*) [44]. During the early stages of aGVHD, the abundance of *Enterococcus* increases, and the abundance of several probiotic strains, such as *Blautia* and *Faecalibacterium*, decreases in the intestines of patients [48]. These changes in the gut microbiota composition further contribute to the pathogenesis of aGVHD through several mechanisms. First, reducing the levels of *Blautia* and *Faecalibacterium* leads to reduced production of microbial metabolites such as SCFAs, which are essential for maintaining intestinal epithelial integrity and preventing aGVHD [49, 50]. Second, *Enterococcus* species, which are typical gram-negative bacteria, release lipopolysaccharides (LPSs) into the bloodstream to activate TLR signaling, leading to NF- κ B activation and the subsequent production of proinflammatory cytokines such as TNF- α , IL-6, and IL-1 β , which drive tissue inflammation and injury. In the present study, we reported that patients with refractory cGVHD also exhibited gut microbiota dysbiosis with overgrowth of *Escherichia-Shigella* and *Enterobacteriaceae*. FMT reduced the abundance of these pathogenic bacteria and increased the abundance of SCFA-producing bacteria (e.g., *Ruminococcus* and *Faecalibacterium*) and *Parabacteroides*. In our study, a fecal microbiota suspension from two different donors was used for FMT. Although analysis of the fecal microbiota suspensions from the two donors revealed that the microbial compositions of Donor 1 and Donor 2 were similar across different taxonomic levels, LDA-based differential analysis revealed that the fecal microbiota suspension from Donor 2 was relatively rich in *Subdoligranulum*, *Ruminococcus*, and *Dorea*, which are well-known SCFA-producing bacteria (Additional file 3: Fig. S2). Among the patients who were treated with the Donor 2 FM mixture, the response rate (62.5%, 5 of 8) was greater than that of the Donor 1 group (25%, 1 of 4), suggesting that SCFA-producing

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Fig. 4 Immunohistochemical staining of various types of immune cells in the colon tissues of patients with refractory disease before and after FMT. Representative images of primary colon tissue samples stained for CD4, CD8, CD11b, CD19, and Foxp3 are shown in the left panel. Scale bars on the left: 200 μ m; scale bars on the right: 50 μ m. The optical density values of the CD4/CD8/CD11b/CD19/Foxp3 proteins in each field were analyzed by Qupath software, and the statistical analysis results of responders and nonresponders before and after FMT are shown in the right panel. FMT fecal microbiota transplantation, Foxp3 Forkhead box protein P3, NR nonresponder; * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, and **** $p < 0.0001$

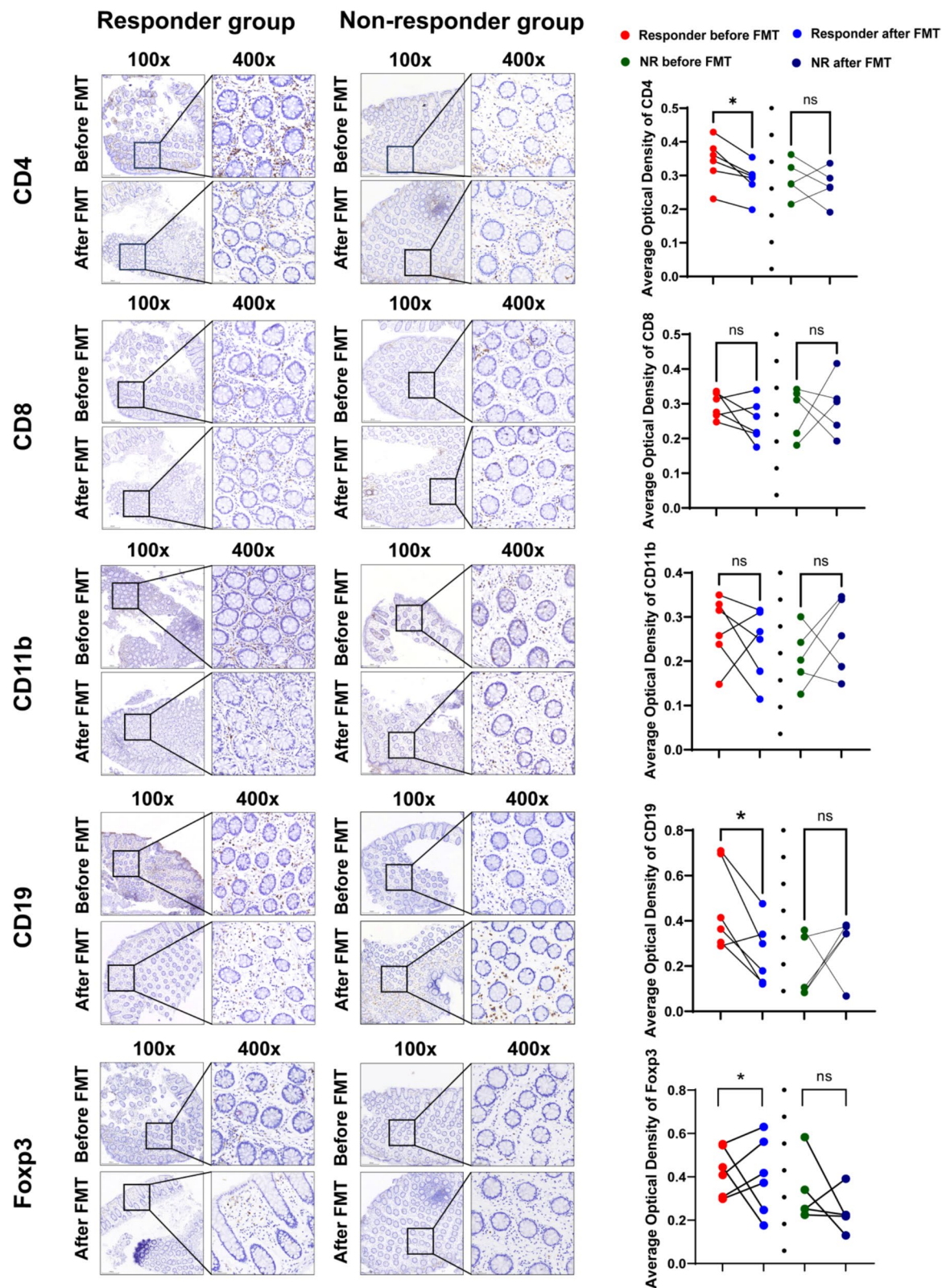


Fig. 4 (See legend on previous page.)

bacteria may contribute to the therapeutic effect of FMT on cGVHD. FMT also significantly increased the number of peripheral $CD4^+CD127^-$ Treg cells, suggesting that an increase in the percentage of Treg cells is critical for suppressing immune overactivation in GVHD. Previous studies have suggested that SCFA levels are decreased in patients with cGVHD [27]. SCFAs increase Treg cell generation, thereby reducing proinflammatory cytokine release and ameliorating immune-mediated tissue damage. In this study, FMT significantly increased the levels of SCFA metabolites, particularly butyric acid, in the feces of responders, which notably increased Treg cell generation in patients with refractory cGVHD (Fig. 3E and Additional file 3: Fig. S4).

Cohousing experiments are widely used as functional methods for investigating microbial transfer between subjects [51–53]. The results of cohousing cGVHD model mice with healthy donor mice suggest that the transfer of fecal microbes from healthy donors to cGVHD model mice alleviated cGVHD, accompanied by increasing levels of SCFAs and Treg cells (Fig. 5). Additionally, colon biopsy analysis of refractory cGVHD patients revealed

marked infiltration of inflammatory cells, including $CD4^-$, $CD8^-$, $CD19^-$, and $CD11b^-$ positive cells, along with severe epithelial damage. After FMT, there was a marked decrease in the infiltration of $CD4^+$ T cells and $CD19^+$ B cells simultaneously in the colonic mucosa, with an increase in Foxp3-positive cells (Fig. 4). Together with increased SCFA levels in the feces of responders, the mechanism of FMT might involve an increase in SCFA levels and the proliferation of Treg cells to diminish B cell infiltration in the colon.

Although our study evaluated the efficacy and safety of FMT treatment in refractory cGVHD patients, some limitations should be considered. First, the study's sample size was small and heterogeneous, which prohibits conclusions on the majority of refractory cGVHD patients. Second, the study had a noncontrolled design. Since a placebo arm was not included, the clinical outcomes should be interpreted with caution. The third limitation is that single-center investigations may not be representative of or generalizable to a wider range of medical centers. Further investigations of FMT as a safe salvage treatment and support from more centers are

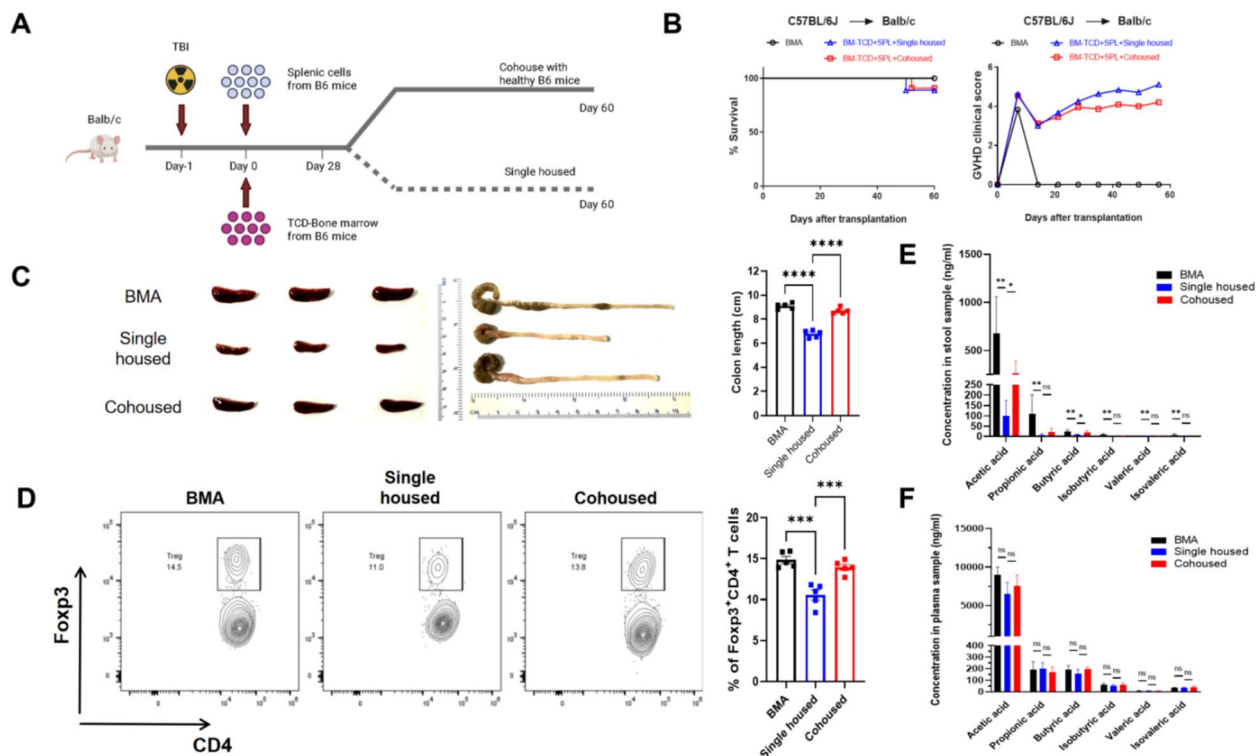


Fig. 5 Cohousing with healthy donor mice reduces the severity of cGVHD via the SCFA-Treg axis. **A** A B6 → BALB/c cGVHD model was established, and the recipients were either housed alone or cohoused with healthy B6 mice. **B** Recipients were monitored for survival and clinical score until 60 days post-BMT ($n = 5$ per group). **C** Pathology of the spleen and colonic organs in the BMA, singly housed and cohoused groups. **D** Expression of Foxp3⁺ cells gated on donor CD4⁺ T cells. **E** SCFA levels were measured in fecal samples from the mice. **F** SCFA levels were measured in plasma samples from the mice. SCFAs short-chain fatty acids, cGVHD chronic GVHD, TBI total body irradiation, TCD T cell depleted, BMA bone marrow alone, Foxp3 Forkhead box protein P3; * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, and **** $p < 0.0001$

needed. Larger clinical trials are needed to explore more patients to evaluate the efficacy of healthy donor FMT in a broader and more diverse population of patients with refractory cGVHD and to determine the optimal route, dose, and timing of donor FMT for these patients.

Conclusions

The results of the present study suggest that FMT is feasible and deserves further investigation for refractory cGVHD patients.

Abbreviations

aGVHD	Acute graft-versus-host disease
ACE	Abundance coverage estimator\
AEs	Adverse events
allo-HSCT	Allogeneic hematopoietic stem cell transplantation
AML	Acute myeloid leukemia
ASV	Amplicon sequence variant
ATG	Antithymocyte globulin
BMA	Bone marrow alone
BM	Bone marrow
BMT	Bone marrow transplantation
BSA	Bovine serum albumin
cGVHD	Chronic graft-versus-host disease
CNIs	Calcineurin inhibitors
CR	Complete response
DAB	Diaminobenzidine
EDTA	Ethylene diamine tetraacetic acid
FDA	U.S. Food and Drug Administration
FMT	Fecal Microbiota Transplantation
Foxp3	Forkhead Box Protein P3
GI tract	Gastrointestinal tract
GVHD	Graft-versus-host disease
HR	Hazard ratio
HRP	Horseradish peroxidase
IL	Interleukin
LC/MS	Liquid chromatography–mass spectrometry
LDA	Linear discriminant analysis
LEfSe	Linear discriminant analysis effect size
LPS	Lipopolysaccharide
MDS	Myelodysplastic syndrome
MMF	Mycophenolate mofetil
MSC	Mesenchymal stem cell
MTX	Methotrexate
NF-κB	Nuclear factor kappa B
NIH	National Institutes of Health
NR	No response
OD	Optical density
ORR	Overall response rate
PBMCs	Peripheral blood mononuclear cell
PBSC	Peripheral blood stem cell
PCoA	Principal coordinate analysis
PR	Partial response
pTreg	Peripheral regulatory T cell
qPCR/PCR	Quantitative polymerase chain reaction/polymerase chain reaction
SCFA	Short-chain fatty acid
SD	Standard deviation
SEM	Standard error of the mean
SLE	Systemic lupus erythematosus
SPF	Specific pathogen-free
ST2	Suppression of tumorigenicity 2
TBI	Total body irradiation
TCD	T cell depleted
TEAE	Treatment-emergent adverse event
Treg	Regulatory T cell

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12916-025-04336-z>.

Additional file 1. CONSORT checklist.

Additional file 2. ARRIVE checklist.

Additional file 3: Supplementary files (Figures S1–S5 & Table S1). Fig. S1-[Pathology checks of colon tissues from the enrolled participants before FMT treatment]. Fig. S2-[Microbiome analysis of the donor fecal microbiota mixture]. Fig. S3-[Intestinal microbial ecosystem analysis in all participants]. Fig. S4-[The effect of short-chain fatty acids on the polarization of CD4⁺ Treg cells in patients with refractory cGVHD]. Fig. S5-[The integrity of colon tissues after FMT treatment]. Table S1-[FMT efficacy for various cGVHD-targeted organs].

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Authors' contributions

LQ, LX and KY designed the study, performed the experiments, analyzed the data, and wrote the manuscript; JD, FH and YS performed the experiments, collected and analyzed the data; YG, NX, ZF, RX, PW, XY, HL, XL, JX, ZW, JS and YC assisted in the research. All the authors read and approved the final manuscript.

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Data availability

The datasets generated and analyzed during the current study are available in the Genome Sequence Archive (GSA: CRA028287) at the China National Genomics Data Center (NGDC) and are publicly accessible at <https://ngdc.cncb.ac.cn/gsa/browse/CRA028287>. Further detailed information and data are available from the corresponding author upon reasonable request. The authors declare no competing interests.

Declarations

Ethics approval and consent to participate

The clinical trials in this study were approved by the Institutional Ethics Committees of the Nanfang Hospital of Southern Medical University (NFEC-202007-K15-01) and are listed at ClinicalTrials.gov (NCT06938165). Written consent was obtained from each participant after providing a full explanation of the purpose and all procedures used in this study.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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